Lab 2 - Sweep - Giant Paper Clip Bookmark

Course: Innovating Product Development with 3D Printing and Additive Manufacturing (TGS-2022017597) Topic 01: Overview of Additive Manufacturing and 3D Scanning TSC objective: A1 - apply the sweep technique to produce a printable path-driven part

Goal

Model a giant paper clip bookmark in Fusion 360 by sweeping a circular profile along a paper-clip path, then slice and 3D print it.

What you'll build

A giant paper clip bookmark modelled with the sweep technique and 3D printed flat.

Tools: Fusion 360, Ultimaker Cura, FDM printer (PLA)

Steps

1. Start a sketch on the XY plane and draw the paper-clip path - three nested U-turns - using the Line and Arc tools.

```
Create Sketch > Line + Arc (3-point) - overall ~90 x 35 mm
```

2. Add fillets so every corner of the path is a smooth arc, then finish the sketch.

```
Sketch > Fillet - radius 5 mm > Finish Sketch
```

3. Create a plane perpendicular to one end of the path and sketch a 2.5 mm circle on it.

```
Construct > Plane Along Path (t=0) > Sketch > Circle Ø 2.5 mm
```

4. Sweep the circle along the path to form the solid clip.

```
Create > Sweep - profile: circle, path: clip sketch
```

5. Export the body as an STL file.

```
File > Export > STL
```

6. Open it in Cura and lay the clip flat on the build plate so no supports are needed.

```
Cura > Open File > clip.stl > Rotate - flat
```

7. Slice with a 0.2 mm profile and print.

```
Cura > Slice > Save to Disk > print
```

Evidence to capture

- Fusion 360 screenshot showing the path, perpendicular profile and completed sweep.
- Cura Preview screenshot confirming full build-plate contact and no supports.
- Photo or inspection note showing that the printed clip flexes and grips pages.

Test it

The preview shows the whole clip printing in contact with the bed; the printed clip flexes slightly and grips a few pages like a real paper clip.